

Homework 1

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Problem 1

Given points x, y, z, t in a metric space (X, d) prove that

$$|d(x, y) - d(z, t)| \leq d(x, z) + d(y, t)$$

Solution. We start with the triangle inequality on d

$$\begin{aligned} d(x, y) &\leq d(x, z) + d(z, y) \\ d(x, y) &\leq d(x, z) + d(z, t) + d(t, y) \\ d(x, y) - d(z, t) &\leq d(x, z) + d(y, t) \end{aligned}$$

Again

$$\begin{aligned} d(z, t) &\leq d(z, x) + d(x, t) \\ d(z, t) &\leq d(z, x) + d(x, y) + d(y, t) \\ d(z, t) - d(x, y) &\leq d(x, z) + d(y, t) \end{aligned}$$

Then we have $a \leq b$ and $-a \leq b$. Thus $|a| \leq b$.
i.e.

$$|d(x, y) - d(z, t)| \leq d(x, z) + d(y, t)$$

□

Problem 2

Suppose that d is a metric for a non-empty set X , and for any $x, y \in X$ define

$$d^{(1)}(x, y) = kd(x, y), \text{ where } k \text{ is a positive constant, } d^{(2)}(x, y) = \min\{1, d(x, y)\}, \quad d^{(3)}(x, y) = \frac{d(x, y)}{1+d(x, y)},$$
$$d^{(4)}(x, y) = d(x, y)^2$$

Prove that $d^{(1)}, d^{(2)}, d^{(3)}$ are metrics for X but $d^{(4)}$ may not be a metric for X

Solution. For each $d^{(i)}$, $i = 1, 2, 3, 4$ it is clear that each function is well defined. Let $x, y, z \in X$. Since we know that d is a metric for X it is clear that $d^{(1)}$ preserves the symmetry and non-negativity since $k > 0$, also $d^{(1)}(x, y) = 0 \iff d(x, y) = 0 \iff x = y$. Then for the triangle inequality, starting with the triangle inequality for d we have the triangle inequality of $d^{(1)}$ once we multiply across by k .

Since $d^{(2)} \leq 0 \iff d \leq 0$, we see it preserves the non-negativity of d and also if $d^{(2)} = 0 \iff d = 0 \iff x = y$, also $d^{(2)}(x, y) = \min\{1, d(x, y)\} = \min\{1, d(y, x)\} = d^{(2)}(y, x)$, so it is symmetric.

Lemma 1. For all $a, b \geq 0$, we have $\min\{1, a + b\} \leq \min\{1, a\} + \min\{1, b\}$

Proof. Case (i). $a + b < 1$

$$a + b \leq \min\{1, a\} + \min\{1, b\} = a + b$$

Since both a and b are less than one, so the min picks them out

Case (ii) $a + b \geq 1$

$$1 \leq \min\{1, a\} + \min\{1, b\}$$

In any selection of a, b this holds. Since if it picks out any of the ones, it is obvious, and the case condition takes care if both a and b get picked □

For the triangle inequality

$$\begin{aligned}
d^{(2)}(x, y) &= \min\{1, d(x, y)\} \\
&\leq \min\{1, d(x, z) + d(z, y)\} \\
&\leq \min\{1, d(x, z)\} + \min\{1, d(z, y)\} \\
&\leq d^{(2)}(x, z) + d^{(2)}(z, y)
\end{aligned}$$

Once again, we see that $d^{(3)} \leq 0 \iff d \leq 0$, which is never the case, thus it is non-negative and also $d^{(3)}(x, y) = 0 \iff d(x, y) = 0 \iff x = y$. Also the symmetry of d implies the symmetry of $d^{(3)}$.

Lemma 2. For $0 < x \leq y$, we have that $1 - \frac{1}{x} \leq 1 - \frac{1}{y}$

Proof. Since $0 < x \leq y \Rightarrow \frac{1}{x} \geq \frac{1}{y} \Rightarrow -\frac{1}{x} \leq -\frac{1}{y} \Rightarrow 1 - \frac{1}{x} \leq 1 - \frac{1}{y}$ □

For the triangle inequality

$$\begin{aligned}
d^{(3)}(x, y) &= \frac{d(x, y)}{1 + d(x, y)} = 1 - \frac{1}{1 + d(x, y)} \\
&\leq 1 - \frac{1}{1 + d(x, z) + d(z, y)} \\
&\leq \frac{d(x, z) + d(z, y)}{1 + d(x, z) + d(z, y)} \\
&\leq \frac{d(x, z)}{1 + d(x, z) + d(z, y)} + \frac{d(z, y)}{1 + d(x, z) + d(z, y)} \\
&\leq \frac{d(x, z)}{1 + d(x, z)} + \frac{d(z, y)}{1 + d(z, y)} \\
&\leq d^{(3)}(x, z) + d^{(3)}(z, y)
\end{aligned}$$

Then for $d^{(4)}$.

Assume it is a metric. Let $X = \mathbb{R}$, and choose $x = 1, z = 1.5, y = 2$

Then $d(x, y) = 1, d(x, z) = 0.5, d(z, y) = 0.5$ with d being the usual Euclidean metric

Looking at the triangle inequality of $d^{(4)}$, we get

$$1 \leq 0.25 + 0.25$$

A contradiction, thus $d^{(4)}$ is not a metric. □

Problem 3

Let (X, d_X) and (Y, d_Y) be metric spaces. For $(x_1, y_1), (x_2, y_2) \in X \times Y$ let

$$\begin{aligned}
d_1((x_1, y_1), (x_2, y_2)) &= d_X(x_1, x_2) + d_Y(y_1, y_2), \\
d_2((x_1, y_1), (x_2, y_2)) &= \sqrt{d_X(x_1, x_2)^2 + d_Y(y_1, y_2)^2}, \\
d_\infty((x_1, y_1), (x_2, y_2)) &= \max\{d_X(x_1, x_2), d_Y(y_1, y_2)\}
\end{aligned}$$

- Prove that for each of d_1, d_2, d_∞ is a metric for $X \times Y$
- Prove that for any $p, q \in X \times Y$

$$d_\infty(p, q) \leq d_2(p, q) \leq d_1(p, q) \leq 2d_\infty(p, q)$$

- Show that the open sets in $X \times Y$ are the same for d_1, d_2, d_∞
- Let U, V be open subsets of X, Y respectively. Show that $U \times V$ is d_i -open in $X \times Y$ for $i = 1, 2, \infty$

Solution. 1.

Clearly each of the metrics d_1, d_2, d_∞ are well defined due to the properties of d_X and d_Y

Let $x_1, x_2, x_3 \in X$ and $y_1, y_2, y_3 \in Y$

Clearly d_1 is non-negative since the sum of two non-neg is non-neg

Also $d_1 = 0 \iff x_1 = x_2$ and $y_1 = y_2$ implying $(x_1, y_1) = (x_2, y_2)$

Clearly d_1 is symmetric, since both d_X and d_Y are

Triangle Inequality

$$\begin{aligned} d_1((x_1, y_1), (x_2, y_2)) &= d_X(x_1, x_2) + d_Y(y_1, y_2) \\ &\leq d_X(x_1, x_3) + d_X(x_3, x_2) + d_Y(y_1, y_3) + d_Y(y_3, y_2) \\ &\leq d_1((x_1, y_1), (x_3, y_3)) + d_1((x_3, y_3), (x_2, y_2)) \end{aligned}$$

Thus d_1 is a metric on $X \times Y$

d_2 is non-negative since $\sqrt{a} \geq 0 \forall a \geq 0$

Similarly $d_2 = 0 \iff$ the square root is 0 $\iff d_X$ and d_Y are zero $\iff$ the pairs are the same

Again due to the symmetry of d_X and d_Y it is clear that d_2 is symmetric

Lemma 3. For $a, b, c, d \geq 0$, we have $\sqrt{(a+b)^2 + (c+d)^2} \leq \sqrt{a^2 + c^2} + \sqrt{b^2 + d^2}$

Proof. Since both sides are non-negative it is equivalent to prove the following

$$(a+b)^2 + (c+d)^2 \leq (a^2 + c^2) + (b^2 + d^2) + 2\sqrt{(a^2 + c^2)(b^2 + d^2)}$$

Then expanding the LHS and cancelling out some terms we get that it is equivalent to proving

$$ab + cd \leq \sqrt{(a^2 + c^2)(b^2 + d^2)}$$

Squaring and expanding

$$a^2b^2 + 2abcd + c^2d^2 \leq a^2b^2 + a^2d^2 + c^2b^2 + c^2d^2$$

$$0 \leq a^2d^2 - 2abcd + c^2b^2$$

$$0 \leq (ad - bc)^2$$

Which is a square and thus always non-negative.

Hence the inequality holds □

For the triangle inequality

$$\begin{aligned} d_2((x_1, y_1), (x_2, y_2)) &= \sqrt{d_X(x_1, x_2)^2 + d_Y(y_1, y_2)^2} \\ &\leq \sqrt{(d_X(x_1, x_3) + d_X(x_3, x_2))^2 + (d_Y(y_1, y_3) + d_Y(y_3, y_2))^2} \\ &\leq \sqrt{d_X(x_1, x_3)^2 + d_Y(y_1, y_3)^2} + \sqrt{d_X(x_3, x_2)^2 + d_Y(y_3, y_2)^2} \\ &\leq d_2((x_1, y_1), (x_3, y_3)) + d_2((x_3, y_3), (x_2, y_2)) \end{aligned}$$

Again, the max of two non-negative numbers is obviously non-negative

Also if the max of two non-neg numbers is 0, then all are 0 implying $(x_1, y_1) = (x_2, y_2)$

Again due to the symmetry of d_X and d_Y it is clear that d_∞ is symmetric

Lemma 4. For all $a, b, c, d \geq 0$, we have $\max\{a+b, c+d\} \leq \max\{a, c\} + \max\{b, d\}$

Proof. Let $M = \max\{a, c\} + \max\{b, d\}$

Since $a \leq \max\{a, c\}$ and $b \leq \max\{b, d\}$, we have $a + b \leq M$

Similarly $c \leq \max\{a, c\}$ and $d \leq \max\{b, d\}$ give $c + d \leq M$

Then $\max\{a+b, c+d\} \leq M$. □

For the triangle inequality, we get

$$\begin{aligned} d_\infty((x_1, y_1), (x_2, y_2)) &= \max\{d_X(x_1, x_2), d_Y(y_1, y_2)\} \\ &\leq \max\{d_X(x_1, x_3) + d_X(x_3, x_2), d_Y(y_1, y_3) + d_Y(y_3, y_2)\} \\ &\leq \max\{d_X(x_1, x_3), d_Y(y_1, y_3)\} + \max\{d_X(x_3, x_2), d_Y(y_3, y_2)\} \\ &\leq d_\infty((x_1, y_1), (x_3, y_3)) + d_\infty((x_3, y_3), (x_2, y_2)) \end{aligned}$$

Thus each of the functions are metrics on $(X \times Y)$ □

Solution. 2.

Let $d_\infty((x_1, y_1), (x_2, y_2)) = \max \{d_X(x_1, x_2), d_Y(y_1, y_2)\} = \alpha$. Then

$$d_2((x_1, y_1), (x_2, y_2)) \geq \sqrt{\alpha^2 + 0^2} = \alpha = d_\infty((x_1, y_1), (x_2, y_2)) \quad (1)$$

Let $d_1((x_1, y_1), (x_2, y_2)) = a + b$, $a, b \geq 0$ Then $d_2((x_1, y_1), (x_2, y_2)) = \sqrt{a^2 + b^2}$

Then

$$\begin{aligned} d_1^2 &= a^2 + b^2 + 2ab, & d_2^2 &= a^2 + b^2 \\ d_1^2 \leq d_2^2 &\Rightarrow d_2((x_1, y_1), (x_2, y_2)) \leq d_1((x_1, y_1), (x_2, y_2)) \end{aligned} \quad (2)$$

Again let $d_\infty = \alpha$. Then $d_1 \leq \alpha + \alpha = 2d_\infty$

$$\Rightarrow d_1((x_1, y_1), (x_2, y_2)) \leq 2d_\infty((x_1, y_1), (x_2, y_2)) \quad (3)$$

Combining 1, 2 and 3 we get, with $p = (x_1, y_1)$ and $q = (x_2, y_2)$

$$d_\infty(p, q) \leq d_2(p, q) \leq d_1(p, q) \leq 2d_\infty(p, q)$$

□

Solution. 3.

Let $U \subset X \times Y$, U is open iff

$$\forall (x_0, y_0) \in U \exists \epsilon \text{ s.t. } B_\epsilon(x_0, y_0) \subset U$$

To prove the open sets are the same under the three different metrics, we must prove that if a set is d_∞ -open then it is also d_1 -open and then also d_2 -open and finally once again to d_∞ completing the cycle.

Let $(x_0, y_0) \in U$

$$\begin{aligned} B_{\epsilon_\infty}^{d_\infty}(x_0, y_0) &= \{(x, y) \in X \times Y \mid \max \{d_X(x, x_0), d_Y(y, y_0)\} < \epsilon_\infty\} \\ B_{\epsilon_1}^{d_1}(x_0, y_0) &= \{(x, y) \in X \times Y \mid d_X(x, x_0) + d_Y(y, y_0) < \epsilon_1\} \\ B_{\epsilon_2}^{d_2}(x_0, y_0) &= \{(x, y) \in X \times Y \mid \sqrt{d_X(x, x_0)^2 + d_Y(y, y_0)^2} < \epsilon_2\} \end{aligned}$$

Assume U is d_∞ -open. i.e. $\exists \epsilon_\infty > 0$ s.t. $B_{\epsilon_\infty}^{d_\infty}(x_0, y_0) \subset U$

Choose $\epsilon_1 = \epsilon_\infty$

Then if $(x, y) \in B_{\epsilon_1}^{d_1}(x_0, y_0) \Rightarrow (x, y) \in B_{\epsilon_\infty}^{d_\infty}(x_0, y_0)$ since $d_\infty \leq d_1$ (from part above) $\Rightarrow B_{\epsilon_1}^{d_1}(x_0, y_0) \subset B_{\epsilon_\infty}^{d_\infty}(x_0, y_0) \subset U$.

Hence U is d_1 -open

Assume U is d_1 -open. i.e. $\exists \epsilon_1 > 0$ s.t. $B_{\epsilon_1}^{d_1}(x_0, y_0) \subset U$

Choose $\epsilon_2 = \frac{1}{2}\epsilon_1$

From the last part we have that $d_1 \leq 2d_2$

So if $(x, y) \in B_{\epsilon_2}^{d_2}(x_0, y_0)$ we get

$$\begin{aligned} &\Rightarrow d_2((x, y), (x_0, y_0)) < \frac{1}{2}\epsilon_1 \\ &\Rightarrow d_1((x, y), (x_0, y_0)) \leq 2d_2((x, y), (x_0, y_0)) < \epsilon_1 \end{aligned}$$

So $(x, y) \in B_{\epsilon_1}^{d_1}(x_0, y_0)$

Hence $B_{\epsilon_2}^{d_2}(x_0, y_0) \subset U$, and thus U is d_2 -open

Assume U is d_2 -open. i.e. $\exists \epsilon_2 > 0$ s.t. $B_{\epsilon_2}^{d_2}(x_0, y_0) \subset U$

Choose $\epsilon_\infty = \frac{1}{2}\epsilon_2$

Again from the last part we have that $d_2 \leq 2d_\infty$

So similar to the last time, if $(x, y) \in B_{\epsilon_\infty}^{d_\infty}(x_0, y_0)$, we get

$$\begin{aligned} d_\infty &< \frac{1}{2}\epsilon_2 \\ &\Rightarrow d_2 \leq 2d_\infty < \epsilon_2 \end{aligned}$$

Then $B_{\epsilon_\infty}^{d_\infty}(x_0, y_0) \subset U$ and hence U is d_∞ -open, completing the cycle.

□

Solution. 4.

Lemma 5. Let $A \subset B, C \subset D$

Then $A \times C \subset B \times D$

Proof.

$$\begin{aligned} A \subset B &\iff a \in A \Rightarrow a \in B \\ C \subset D &\iff c \in C \Rightarrow c \in D \end{aligned}$$

From the properties of the Cartesian Product we get

$$(x, y) \in X \times Y \iff x \in X \text{ and } y \in Y$$

Combining these we get

$$(a, c) \in A \times C \Rightarrow (a, c) \in B \times D$$

Therefore $A \times C \subset B \times D$ □

$U \subset X$ open and $V \subset Y$ open

Thus $\forall x_0 \in U \exists \epsilon_x$ s.t. $B_{\epsilon_x}^{d_X}(x_0) \subset U$ and $\forall y_0 \in V \exists \epsilon_y$ s.t. $B_{\epsilon_y}^{d_Y}(y_0) \subset V$

It suffices to prove that $U \times V \subset X \times Y$ is d_i -open for some $i = 1, 2, \infty$ as a consequence of the last part. I will choose to do it for d_∞

From Lemma 5, we have that $U \times V \subset X \times Y$

$U \times V \subset X \times Y$ is open iff

$$\forall (x_0, y_0) \in U \times V \exists \epsilon_\infty \text{ s.t. } B_{\epsilon_\infty}^{d_\infty}(x_0, y_0) \subset U \times V$$

$$B_{\epsilon_\infty}^{d_\infty}(x_0, y_0) = \{(x, y) \in X \times Y \mid \max\{d_X(x, x_0), d_Y(y, y_0)\} < \epsilon_\infty\}$$

Choosing $\epsilon_\infty = \min\{\epsilon_x, \epsilon_y\}$, we have that $d_X(x, x_0) < \epsilon_x$ and $d_Y(y, y_0) < \epsilon_y$

Meaning if that $(x, y) \in B_{\epsilon_\infty}^{d_\infty}(x_0, y_0) \Rightarrow (x, y) \in B_{\epsilon_x}^{d_X}(x_0) \times B_{\epsilon_y}^{d_Y}(y_0)$

So then $B_{\epsilon_\infty}^{d_\infty}(x_0, y_0) \subset B_{\epsilon_x}^{d_X}(x_0) \times B_{\epsilon_y}^{d_Y}(y_0) \subset U \times V$

Therefore $U \times V$ is d_∞ -open

Hence it is d_i -open for $i = 1, 2, \infty$ □

Problem 4

Let (X, d) be a metric space and consider $X \times X$ as a metric space with the metric d_1 of Problem 3. Show that $d : X \times X \rightarrow \mathbb{R}$ is continuous

$$d_1((x_1, x_2), (x_3, x_4)) = d(x_1, x_3) + d(x_2, x_4)$$

Solution. Let $(x_0, y_0) \in X \times X$

$d : X \times X \rightarrow \mathbb{R}$ is continuous at (x_0, y_0) iff

$$\forall \epsilon > 0 \exists \delta > 0 \text{ s.t. } d_1((x_0, y_0), (x, y)) < \delta \Rightarrow |d(x_0, y_0) - d(x, y)| < \epsilon$$

Let $\epsilon > 0$

Choose $\delta = \epsilon$

Then $d_1((x_0, y_0), (x, y)) < \delta$ gives

$$d(x_0, x) + d(y_0, y) < \epsilon$$

Then using the property from Problem 1, we get

$$|d(x_0, y_0) - d(x, y)| \leq d(x_0, x) + d(y_0, y) < \epsilon$$

Which satisfies the definition of continuity above □

Problem 5

Consider the set of all maps from the natural numbers to the real numbers

$$X = \mathbb{R}^{\mathbb{N}}$$

Define

$$d(x, y) = \sum_{j=1}^{\infty} \frac{1}{2^j} \frac{|x_j - y_j|}{1 + |x_j - y_j|}$$

Show that (X, d) is a metric space

Solution. Let $x, y, z \in X$

First I should check whether or not the metric is well-defined since it is not immediately obvious.

Each term in the series of partial sums of $d(x, y)$ is non-negative since $|x_j - y_j| \geq 0 \forall j$

Also, since $\frac{|x_j - y_j|}{1 + |x_j - y_j|} \leq 1$ for all j , we get

$$0 \leq d(x, y) \leq \sum_{j=1}^{\infty} \frac{1}{2^j}$$

Since, $\sum_{j=1}^{\infty} \frac{1}{2^j}$ converges, so its limit is an upper bound on the series of partial sums, which is also monotone since it is the sum of non-negative numbers.

Hence it is convergent and thus $d(x, y)$ is well defined

It is obvious that $d(x, y)$ is non-negative since $|x_j - y_j| \geq 0 \forall j$

Then if $d(x, y) = 0$ we see that this happens iff $|x_j - y_j| = 0 \forall j$ i.e. $x = y$

Also $|x_j - y_j| = |y_j - x_j|$ and thus $d(x, y) = d(y, x)$

Then for the triangle inequality (using lemma 2 from earlier)

$$\begin{aligned} d(x, y) &= \sum_{j=1}^{\infty} \frac{1}{2^j} \frac{|x_j - y_j|}{1 + |x_j - y_j|} \\ &= \sum_{j=1}^{\infty} \frac{1}{2^j} \left(1 - \frac{1}{1 + |x_j - y_j|} \right) \\ &\leq \sum_{j=1}^{\infty} \frac{1}{2^j} \left(1 - \frac{1}{1 + |x_j - z_j| + |z_j - y_j|} \right) \\ &\leq \sum_{j=1}^{\infty} \frac{1}{2^j} \left(\frac{|x_j - z_j| + |z_j - y_j|}{1 + |x_j - z_j| + |z_j - y_j|} \right) \\ &\leq \sum_{j=1}^{\infty} \frac{1}{2^j} \left(\frac{|x_j - z_j|}{1 + |x_j - z_j| + |z_j - y_j|} + \frac{|z_j - y_j|}{1 + |x_j - z_j| + |z_j - y_j|} \right) \\ &\leq \sum_{j=1}^{\infty} \frac{1}{2^j} \left(\frac{|x_j - z_j|}{1 + |x_j - z_j|} + \frac{|z_j - y_j|}{1 + |z_j - y_j|} \right) \\ &\leq d(x, z) + d(z, y) \end{aligned}$$

Hence (X, d) is a metric space

□